

Disrupt 141-C — v3 handoff (September 2026)

Read this first. v3 re-checks the whole model against the 23-sheet working set and rebuilds it. v1 and v2

files are all still here (`04\_model/viewer\_standalone\_v2.html`, `Disrupt\_141C.glb`, `Disrupt\_141C\_v2\_corrected.glb`).

What was wrong in v2, and what v3 does

| # | Problem found by overlaying the model on A-111 / A-112 / A-113 | v3 fix |

| --- | --- | --- |

| 1 | **First and second floors ~4 ft south of the ground floor** (and ~1 % larger). A-112/A-113 are plotted at a different position than A-111; v1 computed a registration (`reg\_floors.json`) but never applied it. Courtyard columns, stair and reception did not stack. | Both sheets registered to A-111 on 12-14 shared structural columns each (courtyard corners, reception corners, north row). Residual RMS **0.2 ft**. `floor\_reg.json`: FF scale 0.9906, dx +10.3 px, dy -144.5 px; SF scale 0.9899, dx +28.8, dy -123.8. Every extraction runs on the registered sheets. |

| 2 | **Second-floor masjid / prayer area (Block D) missing**. v1 cut the reception rectangle out of the second floor too. | The masjid sits over the double-height reception (A-301: 22'-1" high, capped at +23'-1"). Only the first floor has the void (19'-3" x 21'-5"). |

| 3 | **Ground-floor north-west compound modelled as a closed box**. v1 grew the ground floor under the whole first floor. A-111 shows open compound there; the emergency exits open onto it. | Ground floor = A-111 only (plus ≤ 2 ft registration slivers). The first-floor block over the compound ($\approx 1,540$ sq ft) stands on 17 columns. |

| 4 | Courtyard / reception rectangles measured on the unregistered first floor (reception 3 ft too far south, gap between it and the courtyard). | Re-measured on A-111: courtyard x 74.7-97.5, y 51.0-79.5 (23'-0" x 28'-4"); reception x 74.8-95.3, y 29.6-51.0 (20'-6" x 21'-5"). |

| 5 | **Most new partitions missing** — the "proposed extension wall" (two lines with grey cross-hatch) was never extracted, only cyan existing walls and solid black. | `hatch.py` detects paired parallel lines with hatch between. Bath rooms, store rooms, workstation partitions etc. are now in. |

| 6 | Envelope was a solid band all round the floor plate, so doors in the plan were walls. | External wall only where the plan draws one (checked across the edge, so door widths are kept). Drawn gaps: open doorways with an 8'-0" lintel on the ground floor ($\leq 5'$ -6", and the covered porch), glazed elsewhere. The GF courtyard edge is the glazed screen. |

| 7 | Floor levels simplified ($x < 55 \Rightarrow +13'$ -11" etc.). | `levels.py`: every level from the LEV tags. GF west wing +2'-0" (to the 4'-6" passage), FF +12'-5" / +13'-3" (south-west) / +13'-11" / +16'-11" (north-west), SF +23'-10" / +25'-9". Wall tops follow the floor above. |

| 8 | Compound wall drawn as the 151 x 130 ft sheet window, not the drawn wall; guard post, foyer, porch, generator room, janitorial block missing or wrong; lawn modelled as a room. | Compound x 0.3-149.5, y 7.8-128.9 as drawn; foyer, covered porch (24'-7"), guard post 6'-0" x 6'-0" outside the gate, generator room and janitorial block added; lawn is open ground. |

| 9 | Stairs positioned from the old frame. | Main stair x 84.0-94.6, y 93.0-108.3 (to roof); west stair x 10.8-27.2, y 78.6-88.2 (GF compound $\rightarrow$ FF $\rightarrow$ SF +25'-9"); service stair x 140.2-147.8, y 93.4-107.7. |

| 10 | Room names on FF/SF mostly "Work stations" with no sizes; FF/SF labels in the wrong frame; > 63 rooms overflowed the 6-bit room id. | 151 rooms (GF 67, FF 50, SF 34) with the stated sizes, OCR'd off the registered sheets and hand-checked. Grid now 8-bit room id (plate bit 256, walk bit 512). |

| 11 | 13 rooms unreachable in the walkthrough. | 11 doorways opened automatically where door frames sealed a room (listed in `model\_log\_v3.json`); furniture layout keeps the walker's 0.6 ft clearance. Every room has a route; only the tiny guard W/C and the gen store are sealed as drawn (Fly mode gets in). |

Kept from v2 (client requests): no spiral stair, no parapets on the +12'-5" / +23'-10" roofs.

Files

- `04\_model/viewer\_standalone.html` — **v3 viewer** (offline, model embedded).
- `04\_model/Disrupt\_141C\_v3.glb` / `.obj` — v3 model for Blender / Twinmotion / D5 (feet, Z-up, same object names + `GF\_Columns`, `Guard\_Post`).
- `10\_v3\_rebuild/model\_pipeline/` — `regsheets.py` (sheet registration), `hatch.py` (partitions), `extract\_plans.py`, `plates.py`, `plate\_edits.py` (every hand correction, with its drawing reference), `levels.py`, `build\_v3.py`.
- `10\_v3\_rebuild/viewer/` — walk / room / furniture pipeline and `build\_viewer\_v3.py`.
- `10\_v3\_rebuild/verification/` — the model overlaid on each registered dimension plan, the three building maps, a cutaway and a walk view.

Rebuild order

```
pdftoppm -r 500 -png -f 6 -l 8 <PDF> ... # A-111/112/113
pdftoppm -r 500 -png -f 12 -l 14 <PDF> ... # A-501/502/503
python3 regsheets.py # registered sheets + composed RCP registration
python3 hatch.py ff-07 ff-08 all11-06
python3 extract_plans.py && python3 plates.py && python3 plate_edits.py && python3
build_v3.py
cd viewer/walk_pipeline && python3 smooth.py && python3 grid.py && python3 rooms.py &&
python3 export.py \
    && python3 furnish.py && python3 export.py && python3 write_glb.py
cd .. && python3 build_viewer_v3.py
```

Scripts carry absolute paths from the build session (`/home/claude/work/...`); change them at the top of each file.

Still open (unchanged)

- **Scale**: still calibrated (41.70 px/ft), not dimensioned. One tape measurement of the front elevation settles it.
- **North point** assumed (entrance faces south).
- ~~~Furniture illustrative~~~ → traced from A-401/402/403 in v3.1.
- Some small partitions (WC cubicles, glass screens drawn as single lines) are not extracted.
- The second-floor strip x 37-46 between the south-west rooms is modelled as an open-air corridor (4'-9"); the X on A-113 there is read as open to sky.
- **Asbestos** cement sheet tagged at +35'-3": confirm before any roof work.

v3.1 and v3.2 (September 2026): read this part first

v3.1 is a second accuracy pass plus furniture, people and equipment. v3.2 adds the **Architect view** and **Fly**.

`04\_model/viewer\_standalone.html` is now v3.2. The v3.0 viewer is kept as `viewer\_standalone\_v3.0.html`.

`04\_model/Disrupt\_141C\_v3\_final.glb` is the final building model with the cleaned wall boxes. Furniture and people are not in the GLB: the viewer builds them from `data/furniture.json`.

v3.1: what changed

| # | Found | Fix |

| --- | --- | --- |

| 1 | v3.0 put the plates on a grid stretched to 41.85 px/ft, while the sheets are calibrated at 41.70 px/ft. Everything drifted up to about 1 ft towards the north. The drawing window also stopped at row 205, which cut off the north-west extension. | `frame.py` is the one frame for everything: PXP 41.70, 0.5 ft cells, window x 400-6700, y 80-5625 px. Every hand-read coordinate goes through `T()` / `TR()`. The first-floor block over the NW compound now reaches its drawn north edge. |

| 2 | Grey stipple and text were being read as cyan "existing wall". | Stricter cyan test: `(b>140)&(g>140)&(r<150)&(b-r>50)&(g-r>50)`. |

| 3 | Plate gaps: the GF generator store and service stair, the SF steps to the west block and the SF 4-seat lounge were missing. Two SF plate holes. | Added in `plate\_edits.py` (ADD_TRUE / HOLES_TRUE), each with its drawing reference. |

| 4 | Furniture was illustrative, and some desks and people were in the compound outside the building. | Seating traced from A-401/402/403 (`detect\_furn.py`, then `regfurn.py` → `furn\_detect.json`, `furn\_reg.json`). Desks and tables are grouped from the traced chairs (`furnish4.py`). An item is placed only on a floor plate and on walkable floor. The only things outdoors are the drawn open-air ones: guard post, vending / tea counter, bench. |

| 5 | People and the generator looked like placeholders. | Procedural people (`m\_people.js`): shalwar kameez, hijab and office wear, poses for typing, phone, meeting and standing. Department equipment (`m\_furn.js`, `furnish3.py`): server racks, UPS, gym, board room, gaming room, and prayer rugs facing the qibla (north-west). Generator with canopy, exhaust, fuel tank and cable tray. About 3.7 M triangles in total. |

Checks after v3.2 (`test\_route.py`, `conn.py`):

- 154 rooms. Every room has a walking route except "Main gate & guard post", which is outside by design.
- Reachability failures: 0.

v3.2: Architect view () and Fly () , for checking the building during construction

 **Architect view** is on the header, in Views → Mode, and in the walk bar.

- **Look:** a white model with black linework. Line edges come from a normal and depth post-process.
- **Plan cut:**
 - Ground: +6'-0"
 - First: +20'-5"
 - Second: +29'-9"
 - Full: no cut
 - Auto: follows the floor the camera is on
- **Drawings on the floors:** A-111, A-112 and A-113, registered to the model, are laid on each floor only where that floor exists. The whole A-111 sheet (gate, lawns, ramps) is on the ground.
- **Wall hatch and colours follow the A-111 legend.** Cut walls show their hatch from above.
 - Existing wall: cyan cross-hatch.
 - Proposed extension wall: grey cross-hatch. Solid black new partitions also count here.
 - Envelope not drawn on the plan: dark.
- **Columns** come from the drawings: grey = existing, blue = proposed.
- **Room tags:** room name, stated size and FFL. Tags are thinned out so they don't overlap.
- **Hover read-out:** room, floor, FFL, and the X / Y / height under the cursor. X and Y are in feet in the model frame (the same frame as the GLB).
-  **Measure:** click two points. Hold Shift on the second click to lock to one axis. The result shows the total and its E-W / N-S / up parts, snapped to ½". Check: the reception measures **20'-6"** wall to wall, matching the 20'-6" on A-111.
- **Walking inside in Architect view:** with the cut on Auto, the ceiling is removed just under the slab above. The drawing and its dimensions show on the floor. Measuring works here too: click without dragging.
- **Views:** "Plan (top)" is a true plan, looking straight down. "Plan 3/4" is the architect axonometric.

✈ **Fly** (header button, walk-bar button, or F while walking) works in both Architect and realistic mode.

- **Start:** south of the gate, about 48 ft up.
- **Controls:**
 - W A S D: move
 - drag: look
 - E or Space: up
 - Q or C: down
 - Shift: fast
 - F: land
- **Behaviour:**
 - Speed increases with altitude.
 - The altitude is shown in the location card.
 - In Architect view on Auto: above the roof you get the second-floor plan cut. Flying low inside a floor gives the ceiling-off cut for that floor.

Code: ``10_v3_rebuild/viewer/src/m_arch.js``. It is spliced into ``viewer_app.js`` between ``// @@ARCH_BEGIN`` and ``// @@ARCH_END`` by ``splice_arch.sh``. Its data comes from ``walk_pipeline/arch_prep.py``:

- `arch\_data.json`: wall classes and columns.
- `arch\_under.json`: the underlay JPEGs.

Useful console hooks, used by the test scripts:

- `\_\_api.arch(true)`
- `\_\_api.archCut("FF")`
- `\_\_api.fly()`
- `\_\_api.setView("planTop")`
- `\_\_api.archState()`
- `\_\_api.project(x,y,z)`

v3.2 rebuild order (after the v3 model steps above)

```
cd viewer/walk_pipeline
python3 smooth.py && python3 grid.py && python3 rooms.py && python3 export.py
python3 furnish4.py && python3 export.py && python3 write_glb.py      # traced furniture +
people
python3 arch_prep.py                                              # architect-view data
cd .. && bash splice.sh && bash splice_arch.sh && python3 build_viewer_v3.py
```

`splice.sh` inserts `m\_core/m\_people/m\_furn/m\_build.js`. `viewer\_app.js` in the archive already contains the spliced result.

Still open

- **Scale** is still calibrated at 41.70 px/ft, not dimensioned. Wall-to-wall checks match the drawn dimensions (reception 20'-6"). One tape measurement on site settles it.
- **North point** is assumed: the entrance faces south.
- WC cubicles and single-line glass screens are still not extracted.
- **Asbestos** cement sheet tagged at +35'-3": confirm before any roof work.

v3.3 (September 2026): room-by-room check against A-101 / A-102 / A-103, schedules A-601 / A-602 / A-603

Read this part first. The full table is `10\_v3\_rebuild/v3.3\_room\_check.md` (and `.csv`).

Why rooms looked "missing" on the ground floor

Most of the 10 ground-floor rooms reported missing were already in the data and the 3D model: CCTV room, shower & changing, workshop, procurement office, medical assist, janitorial, female toilet, both lobbies, both huddle hubs, water pumps & tools, sitting.

The building map drew labels biggest room first and **dropped any label that collided** with one already placed, so small rooms lost their names, more so on smaller screens.

v3.3 changes how labels are placed:

- **Building map** (``src/m_map.js``, spliced between ``// @@MAP_BEGIN`` / ``// @@MAP_END`` by ``splice_map.sh``):
 - Every room is labelled. The map tries the drawing's label position, then the room's inner point, shrinking the font to 9 px.
 - A room that still does not fit gets a numbered marker. It is also listed in "Zoom in to read these" (top-left); clicking an entry zooms to the room.
 - Zoom: mouse wheel, pinch, + / - / ↗.
 - Pan: drag.
 - **Drawing** tick box: lays the registered A-111 / A-112 / A-113 under the colours.
- **Search:** finds rooms by name or by size, e.g. ``13'-0"``. It also finds drawing notes (CB, 4S / 6S, exits, counters, voids).
- **Architect view tags:**
 - Every room gets a tag, plus the drawing notes.
 - A tag that doesn't fit falls back to a compact form (name only) or to the room's inner point.
 - "All (overlap)" shows every tag regardless of collisions.
 - The bar shows how many rooms are tagged at the current zoom.
- **Cutaway 3D labels:** shown for every named room larger than 18 sq ft (was 70).

Data changes

- **Anchors** (``walk_pipeline/anchors.py``: ``PATCH``, ``MOVE``, ``A_V33``, ``NOTES``, ``CAP_BY_NAME``):
 - Names and sizes follow the drawings.
 - New rooms added.
 - Drawing notes are exported as ``floors[k].notes`` in ``walkdata.json``.
 - Each room now also carries ``a`` (area), ``p`` (inner label point) and ``b`` (bounding box).
- **Room regions** (``rooms.py``): door-size gaps ($\leq 3'-6"$) between collinear wall runs are closed for the region growth only. A label region now stops at its door; walking is unchanged.
- **Stay lev. 9'-4" × 8'-8"** (A-112, west-stair landing with the fire exit): a landing slab was added to the model.
 - Where: ``plate_edits.py`` `ADD_TRUE` FF.
 - Effect: +81 sq ft FF plate and one more column under it.
- **First-floor glazing** now sits on a 1'-0" upstand with its head at 8'-0". This follows A-602 W1 / W2 / W3 (7'-0" high on 1'-0"). See ``build_v3.py``.
- **Furniture:**
 - Narrow passages (under 60 sq ft) get no plants or standing people.
 - The refreshment tea counter falls back to 4'-6" when 6'-0" doesn't fit.

Where the drawings and the request disagreed (drawing values kept)

Room	Size given in the request	Size on the drawings
---	---	---
Medical assist	15'-6"	19'-4" × 8'-4" (A-101 and A-111)
Lobbies	8'-6", 8'-3"	8'-1" × 6'-9" and 11'-6" × 6'-3"
Huddle hubs	10'-1" × 11'-10", 6'-0" × 13'-6"	19'-7" × 11'-10" and 8'-0" × 13'-5"

- The drawing name is "WATER PUMPS & TOOLS", with "U G W TANK" drawn under it.
- "Male toilet" is kept: A-101 draws the WCs without a name, but A-111 names it MALE TOILET 17'-9" × 16'-7".
- The refreshment area (red note) is a separate open strip east of medical assist. Both are shown.

Door / window schedules

A-601 (ground floor; the sheet itself is numbered A-401)

- Doors: rows sum to 30 = TOTAL 30.
- Windows: 4 = TOTAL 4.
- Against the plan tags on A-101:
 - FG3 4'-0" × 7'-0" is tagged **twice** on A-101 (south of the huddle, and at admin check in / out) but scheduled 01. The plan shows 5 windows, not 4.
 - GDW 7'-6" is listed for "READING ROOM", but A-101 tags GDW on the 8'-3" × 5'-0" huddle by the board room. The reading room carries SDF 16'-0".
 - The gaming-room door is tagged "D1" with the size 3'-6" × 8'-0". That size is D2, and A-601 lists GAME ROOM under D2. This is a tag typo.
- Other counts on A-101 match the schedule:
 - D2: 7 (including the gaming room)
 - D3: 9
 - PVD: 7 (3 changing, 3 commode, janitorial)
 - DD1 / DD2 / DD3: 1 each
 - D1: 2
- Model openings at the A-101 tags were measured automatically (to ½ ft).
 - 10 of 16 are within ±6" of the scheduled width.
 - Still to check:

Door	Location (x, y)	Model opening	Scheduled
---	---	---	---
D2 procurement office	(141, 45)	2'-0" 3'-6"	
D2 admin dining	(136, 92)	4'-6" 3'-6"	
D3 admin / guard washroom	(139, 82)	1'-6" 3'-0"	
D3 gen store	(141, 112)	5'-6" 3'-0"	
DD2 sitting	(78, 108)	10'-6" 6'-0"	
GDW huddle	(109, 30)	6'-6" 7'-6"	

A-602 (first floor)

- The doors table sums to **32 while TOTAL says 23** (a known drawing error).
- Also, GDW 19'-0" is 01 in the table but its elevation note lists meeting room + meeting room = 02.
- Windows: 3 + 2 + 4 = 9 = TOTAL.
- A-102 carries no door tags, so doors can only be checked against the room lists.

A-603 (second floor)

- 2 + 1 + 1 + 2 + 4 + 3 + 10 = 23 = TOTAL.

- There are no windows in the schedule.

Marks are per sheet, not per project. The same mark has different sizes on different floors:

Mark	A-601 (GF)	A-602 (FF)	A-603 (SF)
---	---	---	---
D1	4'-0"	3'-6"	4'-0"
D2	3'-6"	3'-0"	3'-6"
GDW1	—	9'-6"	7'-0"
PVD	2'-3"	2'-0"	2'-0"

- All scheduled doors are 8'-0" high. The model's door head is 8'-0".

Other drawing errors (not model errors): A-000 lists Section A-A as "5-301", which should be A-301.

Still open after v3.3

- The six GF door openings listed above that differ from the schedule.
- The **Stay lev.** landing's south edge follows the double line drawn on A-112 and is modelled as a wall. It may be a railing; confirm on site.
- The landing's 4'-0" double door into the stair enclosure is glazed, like every other first-floor envelope gap.

Checks after v3.3

- `test\_route.py`: 159 rooms. Every room has a walking route except the main gate & guard post, which is outside by design.
- `conn.py` is a stricter 0.75 ft-body check. It flags the gaming room and the new 4'-0" basin passage because their shared door strip is narrow. The viewer's own walker (0.6 ft) reaches both.

v3.3 rebuild order

```
cd model_pipeline && python3 plate_edits.py && python3 build_v3.py      # only if plates /  
walls change  
cd ../viewer/walk_pipeline  
python3 smooth.py && python3 grid.py && python3 rooms.py && python3 export.py  
python3 furnish4.py && python3 export.py && python3 write_glb.py && python3 arch_prep.py  
cd .. && bash splice.sh && bash splice_arch.sh && bash splice_map.sh && python3  
build_viewer_v3.py
```

The key plans were registered onto the dimension sheets with AKAZE + RANSAC (`10\_v3\_rebuild/key\_plan\_check/reg101.py`, `reg\_key.json`; residual under 1 px). The overlays used for the room-by-room read are in the same folder.

v3.4 (September 2026): building map regions, labels and sizes

Only map data changed: room regions, labels and room texts. **No 3D walls, slabs or furniture were touched.**

The model GLB is the v3.3 one, and furniture.json and walkF_*.npy are unchanged.

1. Rooms whose colour now reaches the drawn walls

`walk\_pipeline/anchors.py` `FILL` holds rectangles read on A-111 / A-112 / A-113 in model feet. `rooms.py` gives every floor cell inside them to the room.

- Thin partitions inside a room's outline (cubicle walls) count in its area. The map still draws them as walls: grid bit 1024 marks a wall cell (`export.py`).
- A filled room is exactly its outline. Cells it had grown into outside the outline go back to the nearest other room.

Room	v3.3 region	v3.4 region	Drawing
---	---	---	---
GF Courtyard	260 sq ft	682 sq ft (+4.5 %)	23'-0" × 28'-4" = 652
GF Board room	257 395 (-6.7 %)	23'-6" × 18'-0" = 423	
GF Male toilet	132 277 (-5.8 %)	17'-9" × 16'-7" = 294	
FF Workstation (below the IT room)	57 196 (+2.1 %)	16'-6" × 11'-8" = 192	
SF Male bath room	117 214 (-6.6 %)	13'-2" × 17'-5" = 229	
SF Rooftop cafeteria	764 1249 (-2.3 %)	24'-1" × 53'-1" = 1278	
SF Recreational area (its cells had sat in the cafeteria)	490 488 (-6.4 %)	27'-2" × 19'-2" = 521	

What changed around these rooms:

- The **SF 4'-0" wide passage** is now only the 4'-2" strip on the courtyard side (x 97.3-101.5), 137 sq ft. The rest went to the cafeteria.
- **GF courtyard:** A-111 dimensions the courtyard as 4'-6" + 18'-8" by 5'-9" + 22'-7". That means its 23'-0" × 28'-4" includes the 4'-6" and 5'-9" passages. Both passages are now drawing notes inside the courtyard, not separate rooms.
- **FF "Huddle hub" (no size):** it is a marked table area inside Workstation 16'-6" × 11'-8". It is now a note too.

2. Markers never cover a label

`src/m\_map.js` works in three passes:

1. Every room label is placed where it fits.
2. Rooms left over get a numbered marker in a free spot: inside the room first, then within 66 px with a leader line. If there is no free spot, the room is only listed and never drawn over a label.
3. The drawing notes are placed last.

Testing: every drawn rectangle was checked pairwise on GF / FF / SF at 1440×900, 1100×700 and 900×640, with 0 overlaps (`tests/t\_ovl.py`). Markers also stay clear of the north arrow, the scale bar, the zoom buttons and the list panel. "Female staff common room / prayer area" gets its full label at all three sizes. The "Zoom in to read these" list starts folded below 1300 px wide. Screenshots with the drawing overlay are `verification/v34\_\*.jpg`.

3. Sizes completed

`anchors.py` `SIZE34` holds the completed sizes. Anything with no dimension on the drawing now reads "size not given on drawing":

```
| Room | v3.3 size text | v3.4 size text | Source |
| --- | --- | --- | --- |
| SF Work stations (22 pax) | 12'-11" | 12'-11" × 14'-9" | A-113: 8'-3" + 3'-0" + 3'-6" |
| FF Huddle hub (the one by the 18'-2" × 25'-7" work stations) | 7'-0" | 7'-0" × ≈3'-4" | The
depth figure is hidden under the label on A-112; 3'-4" is measured. 4S. |
| FF Huddle hub (inside Workstation 16'-6" × 11'-8") | — | size not given on drawing | Now a
note |
| GF Entrance porch | 24'-7" wide | 24'-7" × 8'-6" | Plus 3'-8" of steps |
| SF Steps to the west block | — | 12'-1" × 29'-0" | A-113 |
| SF Huddle hub (10 pax) | — | size not given on drawing | A-113 dimensions the whole room as
Work stations 19'-5" × 15'-2" |
| SF 4S booth | — | size not given on drawing | About 7'-4" × 3'-9" measured |
| GF CB booths | 3'-4" wide | 3'-4" × 3'-4" | — |
```

Other rows now marked "size not given on drawing":

- GF: green lawn, open to sky, refreshment area, compound, north-west compound, service stair lobby
- FF: green areas
- SF: fire exit

Passages, whose drawings give only the width, now read "... wide (length not given on drawing)".

Rooms still far from their drawn size

These regions are not in the v3.4 scope. They are left as grown from the drawn door openings; the list is in `v3.4\_region\_check.csv`.

- Some are small booths whose region spills into the space in front of them.
- Some are open-plan areas sharing one space with a neighbour.

The same `FILL` mechanism fixes any of them: add the room's rectangles and rerun `rooms.py`, then `export.py`.

Checks after v3.4

- `test\_route.py`: 156 rooms (3 spaces became notes). Every room has a route except the main gate, which is outside by design.

v3.4 rebuild order (map data only)

```
cd viewer/walk_pipeline && python3 rooms.py && python3 export.py  
cd .. && bash splice_map.sh && bash splice_arch.sh && python3 build_viewer_v3.py
```